CALPUS

Knowledge Base

Domestic Logistics Management

BBA3CJ202 . Third Semester UG . Calicut University (CUFYUGP)

Auto-compiled from 4 study file(s) and 1 question paper(s) in this folder.

Part 1 — Study Material

Logistic Management Semester 3 Module-1

Basics of Logistics and Logistics Management

Unit 1: Logistics Definition

Meaning of Logistics Logistics is fundamentally about the efficient management of flow. It refers to the comprehensive process of planning, implementing, and controlling the movement and storage of goods, services, and related information from the point of origin to the point of final consumption.

In a modern business context, logistics is often defined by the 7 Rs: Delivering the Right Product, to the Right Customer, in the Right Quantity, in the Right Condition, at the Right Place, at the Right Time, and at the Right Cost.

The primary goal is customer satisfaction. Logistics ensures that supply chains remain fluid, allowing products to move seamlessly from suppliers to manufacturers, through warehouses, to retailers, and finally into the hands of the consumer. While traditionally associated with just transportation, modern logistics encompasses a complex web of activities including inventory management, packaging, material handling, and sophisticated information systems.

PRACTICAL EXAMPLE: E-COMMERCE DELIVERY

When you purchase a smartphone online, the logistics cycle involves several precision steps:

- Warehousing: Storing the device in a climate-controlled fulfillment center.
- Fulfillment: Picking and specialized protective packaging.
- Logistics Flow: Transfer via long-haul trucks to local sorting hubs.
- Information Tracking: Real-time GPS data shared with the customer.
- Last-Mile Delivery: The final courier trip to the customer's doorstep.

Unit 2: History and Evolution

Early History of Logistics The roots of logistics are deeply embedded in military strategy. Historically, the success of ancient empires—from Rome to the Mongols—depended on their ability to move food, weapons, clothing, and medical supplies over vast distances to support soldiers in the field. The term itself is derived from military administration, where "Logistikos" referred to the skilled movement and supply of resources.

The Evolution to Business Logistics As the world transitioned through the Industrial Revolution, the need for mass distribution transformed logistics from a military necessity into a commercial powerhouse.

– The Industrial Era: Factories began producing goods in unprecedented volumes, necessitating the development of railways, shipping lanes, and standardized warehousing.

– The Infrastructure Boom: Post-WWII developments in highways and air freight allowed for global trade expansion.

– The Digital Age (Logistics 4.0): Today, logistics is technology-driven. Companies leverage Artificial Intelligence (AI), Internet of Things (IoT), and Robotic Process Automation to predict demand and automate warehouses.

Modern logistics is no longer just a "back-office" transportation activity; it is a strategic function that can provide a significant competitive advantage in the global marketplace.

Unit 3: Objectives of Logistics

The core objective of any logistics system is to bridge the gap between production and consumption efficiently. It aims to maximize service levels while minimizing costs.

Key Strategic Objectives

– Right Product & Timely Delivery: Accuracy is paramount. Delivering the wrong item or arriving late damages brand reputation. In sectors like healthcare or perishables, speed is literally a matter of survival.

– Cost Optimization: By streamlining transportation routes and improving warehouse layouts, businesses can significantly reduce their operational overhead, leading to higher profitability.

– Enhanced Customer Service: In the age of "Amazon Prime" expectations, providing fast delivery, easy tracking, and reliable availability is the primary way to secure customer loyalty.

– Strategic Inventory Management: Balancing the "Cost of Carrying" (storage) vs. the "Cost of Stockouts" (lost sales). Efficient logistics ensures just enough inventory to meet demand without wasting capital on excess stock.

– Operational Continuity: Ensuring a smooth, uninterrupted flow of materials to prevent production bottlenecks and supply chain disruptions.

Unit 4: Elements of Logistics

To understand how logistics works, one must look at its interconnected components. These elements function as a single system to move goods efficiently.

– Transportation: The most visible element. It involves selecting the best mode (Road, Rail, Air, Sea, or Pipeline) based on speed, cost, and the nature of the cargo.

– Warehousing & Storage: Acting as a buffer in the supply chain, warehouses hold goods until they are needed, protecting them from damage and ensuring they are located close to the end-market.

– Inventory Control: The mathematical side of logistics—tracking what is in stock, what is moving, and when to reorder.

– Material Handling: The internal movement of goods using specialized equipment like forklifts, conveyors, and AGVs (Automated Guided Vehicles). Proper handling reduces breakage and labor costs.

– Protective Packaging: Packaging serves a dual purpose: protecting the product during rigorous transit and facilitating easier handling and storage.

– Information Flow: The "central nervous system" of logistics. It involves data exchange regarding orders, stock levels, and shipment tracking, often managed via ERP (Enterprise Resource Planning) systems.

Unit 5: Logistics Management Definition

What is Logistics Management?

Logistics management is the specialized administrative field that plans, implements, and controls the efficient, effective forward and reverse flow and storage of goods. It is the bridge between the point of origin (suppliers) and the point of consumption (customers).

Core Features and Importance Logistics management is characterized by its focus on Integration. It doesn't look at transportation or warehousing in isolation but coordinates them to work as a unified machine. It supports production by ensuring raw materials arrive on time and supports distribution by getting finished goods to the market.

Competitive Advantage: In a crowded market, companies with superior logistics management (like Zara or Apple) can respond to trends faster and keep their prices lower by eliminating waste in their distribution networks.

Unit 6: Process of Logistics Management

The process is a continuous cycle designed to maintain high service levels. It typically follows these stages:

– Order Processing: Capturing the customer's requirement and initiating the fulfillment sequence. Accuracy here prevents errors down the line.

– Inventory Planning: Deciding which warehouse will fulfill the order based on proximity and stock availability.

– Storage & Fulfillment: The physical act of picking the items and preparing them for shipment.

– Distribution Strategy: Choosing the most cost-effective transportation route and carrier to ensure timely arrival.

– Last-Mile Delivery: The final, often most expensive, leg of the journey where the product reaches the end-user.

– Reverse Logistics: The management of returns. Whether due to defects or customer preference, handling returns efficiently is vital for e-commerce sustainability and customer trust.

Unit 7: Role of Logistics in the Global Economy

Logistics is often referred to as the "backbone of the economy." Its health is a direct indicator of a nation's economic prosperity.

Economic Impact Drivers:

– Facilitating Global Trade: Efficient ports and shipping lanes allow countries to export their specialties and import what they lack, boosting GDP.

– Industrial Support: Manufacturing cannot exist without a reliable "Just-in-Time" supply of parts and materials.

– Job Creation: The logistics sector is a massive employer, providing millions of roles in trucking, port operations, software development, and warehouse management.

– Agricultural Stability: Through Cold Chain Logistics, perishable food can be transported across continents, reducing waste and ensuring food security.

– Standard of Living: By reducing the cost of moving goods, logistics makes products more affordable for the general public, thereby increasing purchasing power and quality of life.

End of Module 1

Logistic Management Semster 3 Module-2

Domestic Logistics Operations 8 Main types of Transportation In domestic logistics, choosing the correct transport configuration determines the speed, cost-efficiency, and reliability of a supply chain. Modern fulfillment relies on eight distinct types of transportation models to keep commerce moving fluently.

– 1. Road Transportation (Trucking): The most versatile and widely used mode of domestic transport. It is categorized into Full Truckload (FTL), where a single shipment occupies the entire capacity of a truck, and Less-than-Truckload (LTL), where multiple shipments from different businesses are consolidated into one vehicle.

– 2. Rail Transportation: The backbone of heavy industrial logistics. Rail is designed for high-volume, long-distance movements of bulk commodities such as raw materials, coal, minerals, and agricultural goods over land.

– 3. Air Freight: The premier premium transportation tier. It is reserved for high-value, time-critical, or perishable commodities requiring ultra-fast delivery windows across long domestic distances.

– 4. Maritime & Ocean Shipping: Primarily used for international trade, but domestically critical for coastal nations utilizing deep-water lanes to move massive freight volume between coastal industrial centers.

– 5. Inland Waterway Transportation: Utilizes domestic rivers, canals, and lakes via barges. It represents one of the most cost-effective and environmentally friendly methods for transporting bulk materials internally.

– 6. Pipeline Transportation: A highly specialized infrastructure mode. It features continuous, automated movement exclusively designed for liquid and gaseous cargo such as crude oil, refined petroleum, and natural gas.

– 7. Intermodal Transportation: The utilization of two or more distinct transport modes (e.g., combining rail and road) to move a single shipment. It relies on standardized ISO shipping containers that shift seamlessly between trains, trucks, and ships without unloading the freight.

– 8. Courier, Express, and Parcel (CEP) / Last-Mile Logistics: The fast-evolving direct-to-consumer infrastructure fueled by e-commerce. It uses hyper-local networks, delivery vans, and two-wheelers to handle individual high-velocity parcel deliveries directly to the end customer's doorstep.

9 Road, Rail, Air and Water Transportation To manage domestic logistics effectively, a manager must understand the operational trade-offs across the four traditional pillars of transportation.

Structural Comparison Matrix Transportation Cost Volume Operational Ideal Cargo Profile Speed Mode Structure Capacity Flexibility Road Consumer goods, Medium Moderate Low to Ultra-High electronics, retail, (Highly Medium (Door-to- short-haul freight scalable) door) Rail Bulk commodities, Medium- Low (High High Low vehicles, heavy Low economies (Station-to- machinery, minerals of scale) station) Air Pharmaceuticals, Ultra- High Low Medium- high-tech High Low electronics, fashion, (Airport- emergency parts based) Water Grain, steel, oil, Low Ultra-Low Ultra- Low (Port- scrap metals, heavy High dependent) industrial equipment Core Operational Profiles Road Logistics Advantages: Offers unparalleled door-to-door accessibility, flexible routing, immediate dispatch capabilities, and seamless integration with last-mile networks.

Disadvantages: Subject to traffic congestion, highly sensitive to fuel price fluctuations, limited capacity per unit, and prone to weather delays.

Rail Logistics Advantages: Exceptional energy efficiency over land, highly secure against cargo theft, low carbon footprint, and insulated from highway traffic.

Disadvantages: Fixed infrastructure layout prevents direct-to-consumer delivery, requires secondary trucking for terminal transfers, and operates on rigid timetables.

Air Logistics Advantages: Unmatched velocity reduces inventory holding times, highly secure terminal handling with low damage rates, and ideal for just-in-time manufacturing models.

Disadvantages: Strict dimensional and weight limits, highly vulnerable to atmospheric weather anomalies, and cost-prohibitive for low-margin goods.

Water Logistics (Inland & Coastal) Advantages: Unbeatable fuel efficiency and pricing metric per ton-mile, capable of handling oversized project cargo that cannot fit on roads or rails.

Disadvantages: Prolonged transit times, highly dependent on seasonal water levels, and geographically constrained to natural or artificial waterways.

10 Barriers of Domestic Logistics Operations Even within a single nation's borders, logistics operations face friction points that degrade performance, extend lead times, and inflate operational budgets.

– 1. Infrastructure Deficits & Congestion: Aging highway systems, bottlenecked rail networks, and outdated inland ports create structural delays. Urban traffic congestion significantly increases fuel consumption and limits the efficiency of urban last-mile delivery windows.

– 2. Fragmented Regional Regulations: Domestic logistics often encounters variations in regional policies. Differences in state-specific axle weight limits, variable toll systems, regional safety mandates, and localized entry permits complicate route planning and vehicle deployment.

– 3. Fluctuating Operating Costs: Fuel prices remain highly volatile and represent a substantial percentage of total line-haul expenses. Fleet maintenance, rising driver wages, and increasing insurance premiums add continuous financial strain.

– 4. Technological Disparity: A significant divide persists between top-tier enterprises and smaller local carriers. The lack of standardized digital infrastructure—such as Electronic Data Interchange (EDI), advanced Warehouse Management Systems (WMS), and real-time telematics—among smaller vendors reduces visibility across the supply chain.

– 5. Last-Mile Inefficiencies: The final leg of delivery is historically the most expensive and complex. Factors like restricted delivery hours in commercial zones, lack of unloading bays, missing recipient addresses, and failed delivery attempts add considerable waste to distribution systems.

11 Difference between Domestic and International Logistics Operations While the basic goal of moving an item from point A to point B remains identical, the operational complexity scales exponentially when transitioning from domestic networks to global commerce.

Head-to-Head Comparison Operational Element Domestic Logistics Operations International Logistics Operations Geographical Scope Contained entirely within defined Spans multiple continents, oceans, national boundaries. Short to and sovereign states. Long-haul medium transit lanes. transit lanes.

Regulatory & Uniform legal framework. No Multilayered legal environments.

Customs customs clearances, border check Mandatory customs filings, import/ inspections, or tariff assessments export duties, and compliance with required. global trade bodies.

Documentation Minimal paperwork. Typically Extremely high. Demands Bills of Complexity requires only a standard domestic Lading, Certificates of Origin, Waybill, Invoice, and Packing List. Consular Invoices, Customs Declarations, and Letters of Credit.

Operational Element Domestic Logistics Operations International Logistics Operations Financial & Conducted using a single national Exposed to volatile currency Currency Risk currency. Standardized domestic exchange rate fluctuations, varying payment terms. international banking regulations, and complex hedging protocols.

Transportation Mix Heavily dominated by road trucking Heavily dependent on multimodal and rail logistics networks. combinations of maritime container shipping and air freight corridors.

Supply Chain High predictability. Real-time GPS Intermittent predictability. Freight Visibility tracking and cellular networks offer transitions through numerous cross- unbroken asset visibility. dock facilities, transshipment ports, and distinct regional carriers.

Cultural & Linguistic Shared cultural business norms Multi-language environments, Factors and unified language baseline unique local business etiquettes, simplifies negotiations. and diverse timezone differences require cross-border coordination.

End of Module 2

Logistic Management Semster 3 Module-3

Documentation in Logistics 12 Documenting and Information Flow: Advices, Planning,

FTL, LTL

In modern supply chain ecosystems, the physical movement of goods is entirely dependent on the digital and physical information that precedes, accompanies, and follows it. If the information flow is broken, physical distribution grinds to a halt.

The Dynamics of Information Flow Information in logistics moves in two directions:

– Forward Flow: Includes inventory status, shipping instructions, advanced notices, and delivery schedules moving from the supplier toward the end consumer.

– Backward Flow: Includes purchase orders, tracking requests, service feedback, and return authorizations (reverse logistics) moving from the consumer back to the supplier.

Shipping Advices & Advanced Shipping Notices (ASN) A Shipping Advice is a formal communication sent by a seller to a buyer confirming that a shipment has been dispatched. In digitized supply chains, this has evolved into the Advanced Shipping Notice

(ASN).

How it works: An ASN is sent electronically via Electronic Data Interchange (EDI) or API before the physical delivery truck arrives at the destination warehouse.

Operational Importance: It contains granular details—such as the exact time of departure, estimated time of arrival (ETA), purchase order numbers, SKU counts, batch numbers, and pallet configurations.

This allows the receiving warehouse manager to plan labor, allocate dock doors, and optimize cross-docking schedules in advance, preventing warehouse congestion.

Logistics Planning and Documentation Before a wheel turns, logistics planning systems must generate documentation that optimizes routes and resources. Documentation at the planning phase includes:

– Load Tender Documents: Offers a shipment to a specific carrier, detailing the weight, volume, commodity type, and rate.

– Route Manifests: A master document provided to drivers outlining the sequence of drop-offs, optimized travel paths, and delivery windows to minimize empty miles.

FTL vs. LTL Documentation Flow The administrative burden and operational flow alter significantly depending on the load configuration:

Full Truckload (FTL) Documentation:

Highly streamlined. One truck, one shipper, one destination. A single Bill of Lading (BOL) is generated at the point of origin. The document remains with the driver throughout the direct route and is signed once at the final destination as proof of delivery (POD). The risk of document mismatch or cargo loss is exceptionally low.

Less-than-Truckload (LTL) Documentation:

Highly complex. Multiple shippers, multiple destinations, consolidated into a single vehicle. Every individual shipment within the truck requires its own independent BOL. As the truck moves through intermediate cross-dock facilities, pieces of freight are unloaded, sorted, and re-routed. LTL tracking relies heavily on PRO Numbers (Progressive Routing Numbers) assigned to each shipment. A master Consolidation Manifest links all these individual tracking numbers together. If one document or barcode fails to scan at a cross-dock, a single sub-shipment can become lost, causing an entire branch of the supply chain to stall. LTL also requires routine Re-weigh and Inspection Certificates when weight disparities occur at carrier terminals.

13 Documentation, Road Receipts / Truck Receipts / Way Bills

(RR / LR)

Surface transport documentation serves as a legal contract, a commercial receipt, and a tool for regulatory compliance.

Legal Nature of Lorry Receipts & Railway Receipts The Lorry Receipt (LR) is a critical legal document issued by a domestic road transport carrier upon taking custody of a consignment. It acts as an operational contract between the consignor (shipper) and the transport company.

Core Functions of an LR:

– Acknowledgment of Goods: It serves as definitive proof that the transport company has received the goods from the shipper in a specified condition.

– Contract of Carriage: It outlines the terms and conditions of transportation, including freight charges (whether "Prepaid" or "To-Pay" at destination), liability limitations for damage, and delivery clauses.

– Document of Title: A "Consignee Copy" of the LR often acts as a title document. The carrier will not release the physical cargo at the destination unless the receiver presents the original, stamped LR copy.

Railway Receipt (RR) Similar to the LR, the Railway Receipt (RR) is issued by the rail authority (such as national railway boards) when goods are accepted for rail freight transport.

Operational Distinctions:

– Negotiable vs. Non-Negotiable: An RR can be issued in a negotiable form, meaning the ownership of the goods can be transferred to a third party (such as a bank or an intermediate buyer) while the cargo is physically in transit on the tracks.

– Station-to-Station Handover: Unlike road transport, which handles door-to-door delivery, the RR strictly governs station-to-station movement. The documentation must specify demurrage conditions (fees charged if the receiver takes too long to unload wagons at the rail terminal).

Waybills and e-Waybills A Waybill is a document issued by a carrier giving details and instructions relating to the shipment of a consignment of goods. It travels with the vehicle and shows the names of the consignor, consignee, point of origin, destination, and the route.

The Electronic Waybill (e-Waybill) Paradigm: In modern regulatory environments, the traditional paper waybill has been replaced by the statutory e-Waybill system. Generated online via specialized government portals, it tracks the movement of goods in real time.

– Tax Compliance & Fraud Prevention: It is mandated for any domestic movement of goods exceeding a specific monetary threshold. It binds the commercial invoice data to the vehicle's registration number.

– Validity Caps: e-Waybills feature hard mathematical validity timelines based on the travel distance (e.g., valid for 1 day for every 200 kilometers). If a truck breaks down or faces logistical delays and the e-Waybill expires without being officially extended online, the carrier can face massive statutory penalties and vehicle seizure by enforcement authorities.

14 Consignment Note CMR When road transport transitions from domestic corridors to cross-border international highways, standard domestic Lorry Receipts are legally insufficient. Operations must comply with the CMR Convention.

Understanding CMR The term CMR stands for Convention relative au contrat de transport international de marchandises par route (Convention on the Contract for the International Carriage of Goods by Road). It is a United Nations treaty ratified by over 50 countries across Europe, Asia, and North Africa to standardize the legal frameworks of cross-border road shipping.

The CMR Consignment Note The CMR Consignment Note is the standardized, mandatory document utilized to execute a contract under this international treaty. It uniformizes carrier liabilities, transit documentation, and dispute resolutions.

CMR International Pad Routing Architecture COPY 1 (RED) Sent to the Consignor / Shipper as immediate proof of cargo handover.

COPY 2

(BLUE) Despatched with the vehicle; travels with physical cargo to the final Consignee.

COPY 3

(GREEN) Retained securely by the Carrier as legal verification of the carriage contract.

COPY 4

(BLACK) Managed internally for administrative records, cross-border border clearings, and freight auditing.

Granular Elements of the CMR Note:

– Quadruplicate Copy System: Every CMR pad contains four color-coded carbon copies with distinct destinations to prevent cross-border disputes as illustrated in the schematic flow above.

– Reservations Clause (Box 18): If a carrier notices that the packaging is damaged or the pallet counts are incorrect during loading, they must write explicit reservations in Box 18. If they leave this blank, the law assumes the goods were handed over in perfect condition, making the carrier liable for any damages discovered at destination.

– Strict Liability Caps: The CMR convention establishes strict caps on how much money a carrier can be sued for if goods are damaged. This liability is calculated using SDR (Special Drawing Rights)—a global basket currency index managed by the IMF—limiting payouts to 8.33 SDR per kilogram of gross weight, insulating carriers from catastrophic financial ruin over high-value cargo.

15 Booking, Invoicing & Information Flow The integration of commercial booking and financial invoicing with real-time operational data forms the lifecycle of execution in logistics.

1. The Booking Phase Logistics execution begins when an order triggers a freight request within a Transportation Management System (TMS).

– Spot Bookings: Executed when a business utilizes the immediate open market to procure a truck or

container. Rates fluctuate daily based on sudden supply-demand imbalances.

– Contract Bookings: Pre-negotiated agreements where a carrier guarantees a fixed volume

capacity at a stabilized rate over a set duration (e.g., 6 months).

– Information Output: The booking phase generates a Booking Confirmation Note, which reserves

equipment, schedules pick-up slots at the facility, and alerts warehouse teams to prepare inventory.

2. The Invoicing & Financial Auditing Phase Once a shipment is delivered and the signed Proof of Delivery (POD) is uploaded into the system, the financial transaction triggers.

– The Commercial Invoice: Created by the seller to detail the transaction value of the physical goods

sold to the buyer.

– The Freight Invoice: Created by the logistics carrier billing the shipper for the transportation

services rendered (including baseline line-haul charges, fuel surcharges, accessorial fees, and toll adjustments).

The Freight Audit and "Three-Way Match" Freight billing is notoriously prone to clerical errors due to complex accessorial charges (such as detention fees, lift-gate fees, or unexpected residential delivery fees). Advanced systems execute an automated Three-Way Match to approve a payout:

Purchase Order Bill of Lading / LR Freight Invoice (PO) Operational proof Financial invoice validating physical outlining detailed line- Commercial mandate shipment counts haul charges, stating initial expected handled by transport accessorial tolls, and rates and inventory carrier. fees.

specifications.

If the data matches within an acceptable variance window, payment is released automatically. If it misaligns (e.g., a carrier charges for FTL when the PO stated LTL), the system automatically flags the invoice for manual dispute resolution.

16 GPS –RFID

Modern logistics has evolved from passive tracking (knowing where a shipment was yesterday) to active tracking (knowing exactly what a shipment is doing right now). This real-time visibility is driven by combining GPS and RFID technologies.

Global Positioning System (GPS) in Logistics GPS uses a network of orbit satellites to pinpoint the geographical coordinates of an asset. In logistics, hardware tracking modules are hardwired into truck fleets, trailers, or individual ocean containers.

Operational Depth Beyond Basic Tracking:

– Dynamic Geofencing: Managers draw invisible, digital perimeters around critical waypoints (e.g., a

1-kilometer radius around a distribution center). When a GPS-tracked truck crosses this virtual boundary, it automatically triggers an automated email or API alert to the warehouse dock crew to prepare for immediate unloading.

– Advanced Telematics & Sensor Blends: Modern GPS hardware does more than track position. It

functions as a sensor hub, transmitting real-time operational diagnostics:

Cold-Chain Monitoring: Inside refrigerated trucks, temperature and humidity sensors feed live data to the GPS unit. If the temperature rises above a safe threshold, an alert is sent instantly to prevent cargo spoilage.

Driver Behavior Metrics: Captures abrupt braking, aggressive cornering, and engine idling times to improve fleet safety and fuel efficiency.

Radio Frequency Identification (RFID) Unlike GPS, which tracks long-distance movement via satellites, RFID is a localized wireless technology used to manage assets inside warehouses, shipping yards, and manufacturing facilities without requiring manual optical scanning.

Technical Components:

– RFID Tags: Small microchips attached to products, cartons, or pallets. They are split into:

Passive Tags: Have no internal power source. They remain dormant until energized by the radio waves emitted from an RFID reader. They are cheap, thin, and have a short reading range (up to 10 meters).

Active Tags: Contain an internal battery, allowing them to continuously broadcast a signal over longer distances (up to 100 meters). These are used to track high-value equipment or entire shipping containers inside a port yard.

– RFID Readers (Antennas): Installed at strategic locations, such as warehouse entry and exit dock doors.

The Operational Leap Over Barcodes: Barcodes require line-of-sight scanning, meaning an operator must point a laser directly at an individual label. RFID eliminates this bottleneck entirely. For example, a forklift drives an entire pallet containing 200 individual electronic boxes through an RFID dock door. The reader scans all 200 passive tags simultaneously in a fraction of a second, instantly updating the inventory system without the operator needing to break open the protective shrink-wrap.

The Visibility Synergy: GPS + RFID Working Together To build a seamless, modern logistics operation, companies use both technologies in tandem to cover different environments:

1. Warehouse 2. Transit Corridor 3. Destination Context Center

GPS SATELLITE

TELEMATICS

RFID PORTAL SCANS RFID INTAKE

Micro-Visibility: Macro-Visibility: Live path PORTALS geofencing, transit routing, Captures exact individual Micro-Visibility: Auto- and active telemetry asset loading and authenticates final diagnostics.

inventory departure package collection and events. closes the trace loop.

By linking localized RFID event logging with macro-level GPS tracking, logistics platforms maintain complete end-to-end visibility across the entire distribution network.

End of Module 3

Logisitc Management Semster 3 Module-4

Vehicle Selection in Domestic Logistics 17 Vehicle Selection: Types of vehicles Selecting the appropriate transportation asset is a primary determinant of a supply chain's efficiency, cost baseline, and physical capability. Fleet deployment is categorized by capacity, weight classes, and mechanical configurations.

Classification by Gross Vehicle Weight (GVW) Vehicles are strictly tiered into standard capacity segments to match commercial density profiles:

– Light Commercial Vehicles (LCVs): Typically rated under 3.5 metric tons. These compact units

(vans, mini-trucks) offer exceptional maneuverability in urban cores and are primarily deployed for high-frequency last-mile parcel distribution, retail replenishment, and direct-to-consumer e- commerce fulfillment.

– Medium Commercial Vehicles (MCVs): Ranging from 3.5 to 7.5 metric tons. These flatbed or box

trucks handle regional transport routes, moving goods from regional distribution hubs to secondary market stores.

– Heavy Commercial Vehicles (HCVs): Operating above 7.5 metric tons (extending to 40+ tons multi-

axle freightliners). These assets are built for long-distance, large-volume line-haul corridors, interconnecting major ports, industrial complexes, and primary warehouses.

Structural & Mechanical Configurations The engineering frame of the vehicle dictates its turning capability, route access, and structural safety limits:

Rigid Trucks: The driver's cabin, engine chassis, and structural cargo-carrying body form a single, unbroken structural unit. Rigid designs are structurally robust and commonly utilized for localized delivery networks due to their predictable handling characteristics.

Articulated Vehicles: Consist of an independent tractor unit containing a high-power engine coupled to a detachable semi-trailer via a pivoting fifth-wheel connection. This articulation mechanism grants the vehicle a significantly sharper turning radius relative to its total payload length, making it the global standard for high-volume, cross-country freight routing.

Operational Selection Criteria Logistics managers evaluate fleet composition using several key operational factors:

– Infrastructure Constraints: Bridge weight limitations, overhead clearance restrictions, narrow inner-

city alleys, and legal emissions zones (LEZs).

– Economic Efficiency: Balancing initial capital expenditure (CapEx) against operational running

parameters, such as fuel burn metrics, tire wear cycles, and preventive maintenance intervals.

– Cargo Volume & Nature: Aligning vehicle payload capacities with customer order sizes to minimize

under-utilized transit runs.

18 Load types and characteristics A vehicle cannot be chosen effectively without a detailed analysis of the cargo load. Cargo parameters govern space management inside the vehicle body and dictate safety compliance metrics.

Physical States & Handling Modalities

– Dry Bulk Cargo: Homogeneous commodities shipped loose and unpackaged (e.g., grains, minerals, coal, cement). These require specialized self-discharging hoppers, tipper bodies, or pneumatic suction setups.

– Liquid & Gaseous Bulk: High-fluidity chemical compounds, petroleum products, liquid food components, or compressed natural gases. These demand pressurized, insulated tank systems with surge-damping internal architectures.

– Unitized & Palletized Cargo: Standard commercial freight secured on uniform wood or plastic pallets (e.g., ISO standard pallets). This configuration enables rapid mechanical cross-docking using standard forklifts, optimizing loading cycles.

Operational Risk Classes Specialized handling protocols must be applied based on the cargo's risk profile:

– Perishables: Temperature-sensitive agricultural produce, dairy items, frozen proteins, and biopharmaceuticals. These require an unbroken cold chain with continuous multi-zone temperature tracking.

– Hazardous Materials (HAZMAT): Explosive, flammable, toxic, or corrosive chemicals. Transport requires legally certified drivers, placard signaling systems, secure segregation matrices, and emergency response kits.

– Fragile & High-Value: Consumer electronics, designer goods, or precision industrial instrumentation. These require high-security locking mechanisms, unmarked trailers, and air-ride suspension setups to mitigate road vibration damage.

The Volumetric Trade-Off: Weighing Out vs. Cubing Out Every transport asset features two strict engineering limits: a maximum weight capacity (payload tonnage) and a maximum spatial capacity (cubic volume). This creates two distinct operational challenges:

1. Weighing Out (High-Density Cargo): Occurs when heavy materials (e.g., steel coils, concrete structures, marble blocks) hit the legal or structural gross vehicle weight capacity while leaving a significant portion of the trailer's physical space empty. Optimization relies on maximizing axle weight distribution.

2. Cubing Out (Low-Density Cargo): Occurs when lightweight, high-volume products (e.g., retail pet clothing, apparel, plastic packaging, insulation sheets) completely fill the physical space of the trailer before reaching the vehicle's legal weight capacity. Optimization focuses on maximizing the trailer's cubic capacity or using double-deck trailer systems.

19 Main types of vehicle body The vehicle's body configuration represents the specialized enclosure engineered to protect, contain, and ease the loading/unloading of specific cargo types.

Primary Industrial Body Type Structural Engineering Profile Applications Flatbed / Platform An open chassis flat platform lacking fixed side Construction steel, walls or an overhead roof structure. Facilitates structural lumber, oversized unrestricted, rapid multi-angle loading using industrial machinery, heavy cranes or forklifts. Cargo must be concrete pipes, and project manually secured using tension straps or cargo.

chains.

Box Trailer / Dry A completely enclosed, rigid rectangular Palletized consumer goods, Van structure manufactured from aluminum or fast-moving consumer fiberglass panels. Offers absolute goods (FMCG), electronics, environmental protection from rain, wind, and apparel, and dry foodstuffs.

sun, alongside high anti-theft security via sealable rear locking rods.

Reefer An advanced insulated box body coupled to a Chilled and frozen foods, (Refrigerated) diesel-powered, automated climate control unit. seafood, dairy products, Features internal thermal air-circulation tracking biological vaccines, and channels to maintain strict multi-zone temperature-sensitive temperatures ranging from -30°C to +20°C. chemicals.

Curtainsider Combines a rigid roof frame and solid rear Palletized industrial freight, doors with structural, high-tension PVC side automotive components, curtains that slide on overhead roller tracks. multi-drop drink distribution, Blends the full structural weather protection of and building materials.

a dry van with the side-loading speed of a flatbed.

Tanker A cylindrical, aerodynamic storage vessel Petroleum products, manufactured from stainless steel or reinforced aviation fuels, liquid alloys. Contains internal transverse baffles chemicals, liquid milk, (bulkheads) designed to slow the momentum of edible oils, and liquefied liquid sloshing, ensuring vehicle stability during gases.

cornering.

Primary Industrial Body Type Structural Engineering Profile Applications Tipper / Dump An open-top, high-strength steel box body Sand, gravel, mining ore, Body equipped with heavy-duty front-mount or under- demolition debris, body hydraulic pistons. Lifts the front of the bed agricultural waste, and bulk to dump loose bulk cargo via gravity through a raw aggregates.

hinged tailgate.

20 Transport resource requirements, Vehicle routing and scheduling issues Once vehicles and loads are defined, logistics execution shifts to mathematical optimization—matching fleet capacity, operational times, and driver availability against real-world delivery constraints.

Transport Resource Requirements To establish a cost-efficient fleet operation, managers must balance three key resources:

– Fleet Capacity Sizing: Determining the optimal mix of owned assets (Core Fleet) vs. spot-market sub-contracted carriers to handle seasonal demand peaks without carrying expensive excess capacity during low cycles.

– Driver Allocation & Compliance: Balancing driver schedules against strict statutory mandates, such as Hours of Service (HOS) laws and mandatory rest periods to prevent operator fatigue.

– Fuel and Maintenance Asset Kits: Structuring preventive maintenance schedules based on telemetry mileage data to prevent roadside breakdowns during customer delivery runs.

The Vehicle Routing Problem (VRP) The **Vehicle Routing Problem** is a central mathematical challenge in logistics. The goal is to determine the most cost-effective set of routes for a fleet of vehicles to service a dispersed group of customers, starting and ending at a central depot.

Capacitated VRP (CVRP): Routes must be mathematically optimized to ensure the total cargo demand assigned to a single pickup run never exceeds the vehicle's legal weight or cubic capacity limits.

VRP with Time Windows (VRPTW): Adds the real-world constraint that deliveries must occur within specific time frames designated by the customer (e.g., a retail mall receiving dock open only from 06:00 AM to 09:00 AM). Missing these windows results in rejected shipments and financial penalties.

Key Routing & Scheduling Operational Bottlenecks Real-world logistics networks must continuously mitigate several operational challenges:

Urban Congestion & The Backhaul Dilemma Dynamic Exceptions Delays Running a delivery truck Sudden vehicle break-downs, empty on its return trip to the abrupt weather storms, or Unpredictable traffic jams warehouse is known as customer order cancellations increase fuel burn and disrupt "Deadheading". This require agile automated tight delivery schedules.

represents a significant waste scheduling systems to re- Mitigation requires real-time of resource capacity. allocate cargo instantly.

dynamic rerouting via GPS Managers must actively telematics.

procure reverse logistics loads or return assets to maintain profitability.

End of Module 4

Part 2 — Question Papers

Question Paper - 2025 (with answer key)

QP Code: D133858 Total Pages: 1 Name:

Register No.

THIRD SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2025

(CUFYUGP)

BBA3CJ201 DOMESTIC LOGISTIC MANAGEMENT

2024 Admission onwards Maximum Time: 2 Hours Maximum Marks: 70

Section A

All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks) 1 What is meant by logistics planning?

2 What do you mean by vehicle routing and scheduling?

3 What is meant by International Logistics Operations?

4 Define Way Bill.

5 What is a Reefer (refrigerated) body type used for?

6 Define Logistics Management.

7 What do you mean by vehicle acquisition?

8 Define GPS and its role in logistics.

9 What are Driver's Hours regulations?

10 How is the cost per kilometer calculated?

Section B

All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks) 11 Differentiate domestic and international logistics operations.

12 Explain the importance of effective Fleet Management.

13 Write notes on standing costs, running costs, and overhead costs.

14 Discuss the role of information flow in logistics coordination.

15 Discuss the critical transport resource requirements necessary for a domestic logistics operation.

16 Explain the importance of planning and resourcing in logistics operations.

17 Compare manual and computer methods of vehicle routing and scheduling.

18 What are the various types of vehicle body?

Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks) Discuss the role of Driver licensing and Driver's Hours regulations in ensuring road safety and legal compliance.

Explain the process of vehicle selection. What factors are considered for choosing different types of vehicles for various operations?

BBA3CJ201 DOMESTIC LOGISTICS MANAGEMENT

ANSWER KEY-SET 2

Section A

Logistics planning is the process of organizing, coordinating, and optimizing the flow of goods, 1 information, and resources from origin to destination. It ensures timely deliveries, efficient resource utilization, and cost-effective operations.

Vehicle routing and scheduling involves planning the routes and timings of vehicles to deliver 2 goods efficiently. It aims to reduce travel time, minimize fuel costs, and ensure timely deliveries.

International logistics operations involve the management and movement of goods across 3 international borders. It includes customs compliance, multi-modal transport, higher documentation requirements, and coordination between multiple countries.

A Way Bill is a transport document that provides details of the shipment, including sender, 4 receiver, and goods information. It helps track the goods in transit but does not confer ownership.

A Reefer body is used to transport perishable goods like food, pharmaceuticals, and chemicals.

It maintains controlled temperatures to preserve product quality during transit.

Logistics Management involves planning, implementing, and controlling the efficient flow and 6 storage of goods, services, and information. Its goal is to meet customer requirements while optimizing costs and resources.

Vehicle acquisition refers to the process of procuring vehicles for a fleet. This can include buying, leasing, or renting vehicles to meet operational requirements.

GPS (Global Positioning System) is a satellite-based navigation system used to track vehicle locations. It helps in route optimization, real-time tracking, and improving delivery efficiency.

Driver's Hours regulations set limits on the number of hours a driver can operate a vehicle.

They ensure driver safety, prevent fatigue, and comply with legal requirements.

Cost per kilometer is calculated by dividing total vehicle operating costs (fuel, maintenance, 10 depreciation, and overheads) by the distance traveled. It helps determine the cost efficiency of transport operations.

SECTION B

Domestic: Operates within a country, involves local regulations, shorter delivery times, and lower costs.

11 International: Cross-border operations, higher documentation and compliance requirements, multi-modal transport, currency and customs management, longer transit times, and higher costs.

Effective fleet management ensures optimal vehicle utilization, reduced downtime, timely 12 maintenance, cost control, and safety compliance. It improves delivery performance and minimizes operational risks.

Standing Costs: Fixed costs like vehicle depreciation, insurance, and loan repayments.

Running Costs: Variable costs incurred during operation, such as fuel, maintenance, and tires.

Overhead Costs: Indirect costs including administrative expenses, driver salaries, and office expenses. Proper management ensures cost efficiency.

Information flow ensures real-time communication among suppliers, transporters, and 14 customers. It supports tracking, inventory management, timely decision-making, and improves coordination to prevent delays and inefficiencies.

Key requirements include reliable vehicles, trained drivers, fuel and maintenance resources, 15 scheduling tools, warehouse facilities, and communication systems. Adequate resource planning ensures smooth operation, timely delivery, and cost efficiency.

Planning and resourcing ensure the right vehicles, personnel, and equipment are available to 16 meet demand. It minimizes delays, reduces costs, optimizes resource utilization, and maintains service quality.

Manual Method: Relies on human judgment, physical records, and calculations; prone to errors and time-consuming.

Computer Method: Uses software for optimization, real-time updates, and automated scheduling; improves accuracy and efficiency.

Differences: Speed, accuracy, adaptability, cost, data handling, and monitoring.

Common types include: Open Body for general goods, Closed Body for secure cargo, Reefer Body for perishable goods, Tanker Body for liquids, Flatbed for oversized loads, and Containerized Body for standardized shipments. Vehicle choice depends on cargo type and transport conditions.

SECTION C

Driver licensing ensures that only trained and qualified individuals operate vehicles, reducing accidents and improving road safety. Driver's Hours regulations prevent fatigue by limiting driving hours and mandating rest periods. Together, these measures maintain legal compliance, enhance operational safety, and protect both drivers and goods from avoidable risks.

Vehicle selection involves identifying operational requirements, analyzing load type, distance, terrain, and delivery frequency. Factors include vehicle capacity, fuel efficiency, maintenance 20 cost, reliability, safety features, compatibility with existing fleet, and total cost of ownership.

Proper selection ensures optimal performance, cost efficiency, and suitability for specific domestic or international operations.

Part 3 — Official Syllabus

Syllabus - Domestic Logistics Management (BBA3CJ202)

ni.evileerged.www BBA Semester III Calicut University Domestic Logistics Management Course Code: BBA3CJ202 – Module SYLLABUS Notes ni.evileerged.www Course Syllabus Course Code: BBA3CJ202 | Course Title: Domestic Logistics Management | Semester: III

Module I: Introduction to Logistics & Supply Chain

- Unit 1: Concept, Scope, and Role of Domestic Logistics in Business.
- Unit 2: Key Components – Inbound, Outbound, and In-plant Logistics.
- Unit 3: Customer Service in Logistics & Order Processing Cycle.
- Unit 4: Logistics System Design and Integrated Supply Chain Management.

Module II: Transportation & Distribution Networks

- Unit 5: Modes of Domestic Transportation – Road, Rail, Air, Inland Waterways,

Pipeline.

- Unit 6: Transport Economics, Cost Factors, Freight Rating & Carrier Selection.
- Unit 7: Multimodal Transportation, Intermodal Freight, and Fleet Management.
- Unit 8: Distribution Network Design & Hub-and-Spoke Architecture.

Module III: Warehousing & Material Handling

- Unit 9: Warehousing Concepts, Functions, Types (Public, Private, Bonded).
- Unit 10: Warehouse Layout, Operations, Pick-Pack-Ship Management.

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- Unit 11: Material Handling Equipment, Automated Storage & Retrieval Systems

(ASRS).

- Unit 12: Packaging for Domestic Logistics, Unitization, Palletization, Containerization.

Module IV: Logistics Technology & Performance Control

- Unit 13: Inventory Management in Logistics – Safety Stock, EOQ, Reorder Point.
- Unit 14: IT in Logistics – Transportation Management Systems (TMS), WMS, RFID, GPS.
- Unit 15: Third-Party Logistics (3PL) & Fourth-Party Logistics (4PL) Providers.
- Unit 16: Reverse Logistics, Green Logistics, and Domestic Logistics Cost Controls.

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